

guage model in speech recognition (Hannun et al., 2014; Xiong et al., 2016), or with camera information for object detection (Kendall & Cipolla, 2016).

In 2005, Niculescu-Mizil & Caruana (2005) showed that neural networks typically produce well-calibrated probabilities on binary classification tasks. While neural networks today are undoubtedly more accurate than they were a decade ago, we discover with great surprise that *modern neural networks are no longer well-calibrated*. This is visualized in Figure 1, which compares a 5-layer LeNet (left) (LeCun et al., 1998) with a 110-layer ResNet (right) (He et al., 2016) on the CIFAR-100 dataset. The top row shows the distribution of prediction confidence (i.e. probabilities associated with the predicted label) as histograms. The average confidence of LeNet closely matches its accuracy, while the average confidence of the ResNet is substantially higher than its accuracy. This is further illustrated in the bottom row reliability diagrams (DeGroot & Fienberg, 1983; Niculescu-Mizil & Caruana, 2005), which show accuracy as a function of confidence. We see that LeNet is well-calibrated, as confidence closely approximates the expected accuracy (i.e. the bars align roughly along the diagonal). On the other hand, the ResNet’s accuracy is better, but does not match its confidence.

Our goal is not only to understand why neural networks have become miscalibrated, but also to identify what methods can alleviate this problem. In this paper, we demonstrate on several computer vision and NLP tasks that neural networks produce confidences that do not represent true probabilities. Additionally, we offer insight and intuition into network training and architectural trends that may cause miscalibration. Finally, we compare various post-processing calibration methods on state-of-the-art neural networks, and introduce several extensions of our own. Surprisingly, we find that a single-parameter variant of Platt scaling (Platt et al., 1999) – which we refer to as *temperature scaling* – is often the most effective method at obtaining calibrated probabilities. Because this method is straightforward to implement with existing deep learning frameworks, it can be easily adopted in practical settings.

2. Definitions

The problem we address in this paper is supervised multi-class classification with neural networks. The input $X \in \mathcal{X}$ and label $Y \in \mathcal{Y} = \{1, \dots, K\}$ are random variables that follow a ground truth joint distribution $\pi(X, Y) = \pi(Y|X)\pi(X)$. Let h be a neural network with $h(X) = (\hat{Y}, \hat{P})$, where $\hat{Y}$ is a class prediction and $\hat{P}$ is its associated confidence, i.e. probability of correctness. We would like the confidence estimate $\hat{P}$ to be calibrated, which intuitively means that $\hat{P}$ represents a true probability. For example, given 100 predictions, each with confidence of

0.8, we expect that 80 should be correctly classified. More formally, we define *perfect calibration* as

$$\mathbb{P}(\hat{Y} = Y \mid \hat{P} = p) = p, \quad \forall p \in [0, 1] \quad (1)$$

where the probability is over the joint distribution. In all practical settings, achieving perfect calibration is impossible. Additionally, the probability in (1) cannot be computed using finitely many samples since $\hat{P}$ is a continuous random variable. This motivates the need for empirical approximations that capture the essence of (1).

Reliability Diagrams (e.g. Figure 1 bottom) are a visual representation of model calibration (DeGroot & Fienberg, 1983; Niculescu-Mizil & Caruana, 2005). These diagrams plot expected sample accuracy as a function of confidence. If the model is perfectly calibrated – i.e. if (1) holds – then the diagram should plot the identity function. Any deviation from a perfect diagonal represents miscalibration.

To estimate the expected accuracy from finite samples, we group predictions into M interval bins (each of size $1/M$) and calculate the accuracy of each bin. Let B_m be the set of indices of samples whose prediction confidence falls into the interval $I_m = (\frac{m-1}{M}, \frac{m}{M}]$. The accuracy of B_m is

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(\hat{y}_i = y_i),$$

where $\hat{y}_i$ and y_i are the predicted and true class labels for sample i . Basic probability tells us that $\text{acc}(B_m)$ is an unbiased and consistent estimator of $\mathbb{P}(\hat{Y} = Y \mid \hat{P} \in I_m)$. We define the average confidence within bin B_m as

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i,$$

where $\hat{p}_i$ is the confidence for sample i . $\text{acc}(B_m)$ and $\text{conf}(B_m)$ approximate the left-hand and right-hand sides of (1) respectively for bin B_m . Therefore, a perfectly calibrated model will have $\text{acc}(B_m) = \text{conf}(B_m)$ for all $m \in \{1, \dots, M\}$. Note that reliability diagrams do not display the proportion of samples in a given bin, and thus cannot be used to estimate how many samples are calibrated.

Expected Calibration Error (ECE). While reliability diagrams are useful visual tools, it is more convenient to have a scalar summary statistic of calibration. Since statistics comparing two distributions cannot be comprehensive, previous works have proposed variants, each with a unique emphasis. One notion of miscalibration is the difference in expectation between confidence and accuracy, i.e.

$$\mathbb{E}_{\hat{P}} \left[\left| \mathbb{P}(\hat{Y} = Y \mid \hat{P} = p) - p \right| \right] \quad (2)$$

Expected Calibration Error (Naeini et al., 2015) – or ECE – approximates (2) by partitioning predictions into M equally-spaced bins (similar to the reliability diagrams) and